

Enterprise Infrastructure Planning for a Startup Company

Course: ITEP 414 – System Administration and Maintenance

Project: Week 2 Portfolio Project

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BSIT- 4B

PART 1 — COMPANY PROFILE

Company Name: Ewan Co.

Nature of Business

Ewan Co. is a startup BPO company that provides customer support, virtual assistance, administrative solutions, and back-office support for small and medium-sized businesses. It aims to help clients reduce operational costs while maintaining high-quality customer interactions and business processes.

Company Vision

To become a leading provider of reliable and innovative business support services that help organizations streamline operations, reduce costs, and achieve sustainable growth through exceptional customer service.

Company Mission

To deliver high-quality and efficient business process solutions through skilled professionals, modern technology, client satisfaction, professionalism, security, and continuous improvement.

Office Location: Unit 8, Ewan Business Hub, Quezon City, Metro Manila, Philippines. **Organizational**

Structure



Employee Distribution

Department	Employees	Main Function
Information Technology	5	Technical support, systems administration, network management
Human Resources	4	Recruitment, employee records, payroll coordination
Finance	5	Accounting, budgeting, financial records management
Operations/Sales	6	Client acquisition, account management, customer relations
TOTAL	20	Employee distribution follows the Week 2 module scenario.

PART 2 — ENTERPRISE HARDWARE INVENTORY

The hardware plan provides one desktop workstation per employee plus shared infrastructure for server operations, networking, printing, storage, and backup. IT and management laptops support mobility, client presentations, and incident response.

Asset ID	Hardware	Qty.	Department	Purpose
HW-001	Desktop Computer	5	IT	Technical support and system administration
HW-002	Desktop Computer	4	HR	Recruitment and employee records management
HW-003	Desktop Computer	5	Finance	Accounting and financial management
HW-004	Desktop Computer	6	Operations/Sales	Client management, sales tracking, and customer support
HW-005	Laptop	5	IT	Mobile on-site support, meetings, and remote troubleshooting
HW-006	Laptop	2	Management	Administrative work and client presentations
HW-007	Server	1	IT	Centralized internal services, file and print services
HW-008	Router	1	Network	Internet connectivity and traffic routing
HW-009	Managed Switch	1	Network	Wired connectivity; VLANs; room for expansion
HW-010	Network Printer	2	Shared	Departmental document printing and scanning
HW-011	UPS	4	IT/Network	Power protection and safe shutdown for critical equipment
HW-012	Wireless Access Point	3	Network	Reliable Wi-Fi coverage throughout one office floor
HW-013	NAS Storage	1	IT	Centralized storage and local backup target
HW-014	External Backup Drive	2	IT	Rotating offline/off-site backup copies
HW-015	Monitor	22	All departments	Employee displays and two management/IT spares

Hardware Justification

20 desktops cover the required employee distribution. Five IT laptops support mobility for troubleshooting and meetings; two management laptops support administration and presentations. The server centralizes internal services, while the NAS and external drives support recovery. UPS units protect the server, NAS, firewall/router, and switch; three APs are proposed to cover an office floor subject to a post-installation Wi-Fi site survey.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Primary operating system for employee workstations
Ubuntu Server	Current LTS	Open source	Server operating system for internal services
Microsoft Office	Microsoft 365 / Current Subscription	Commercial	Documents, spreadsheets, email, and presentations
Visual Studio Code	Current stable	Free	Code editing and automation/configuration work for IT
Git	Current stable	Open source	Version control for scripts and internal development
GitHub Desktop	Current version	Free	Graphical Git and GitHub repository management
VirtualBox	Current version	Open source	VM testing and training without affecting production
Google Chrome	Current stable	Free	Web browsing and web-application testing
Microsoft Defender	Built-in/current	Included	Endpoint malware protection and security controls
AnyDesk	Current version	Commercial/free options	Authorized remote technical support only
7-Zip	Current stable	Open source	File compression and archive management

Software Justification

Windows 11 Pro provides a familiar, business-ready workstation environment. Ubuntu Server provides an open-source platform for internal services. Microsoft 365 supports documentation and administration. VS Code, Git, GitHub Desktop, and VirtualBox support IT scripting, configuration, and testing. Chrome supports browser-based tools. Defender protects endpoints, AnyDesk enables controlled remote support, and 7-Zip manages archives. Maintain an approved software list, apply updates, and record license assignments.

PART 4 — ENTERPRISE NETWORK INVENTORY

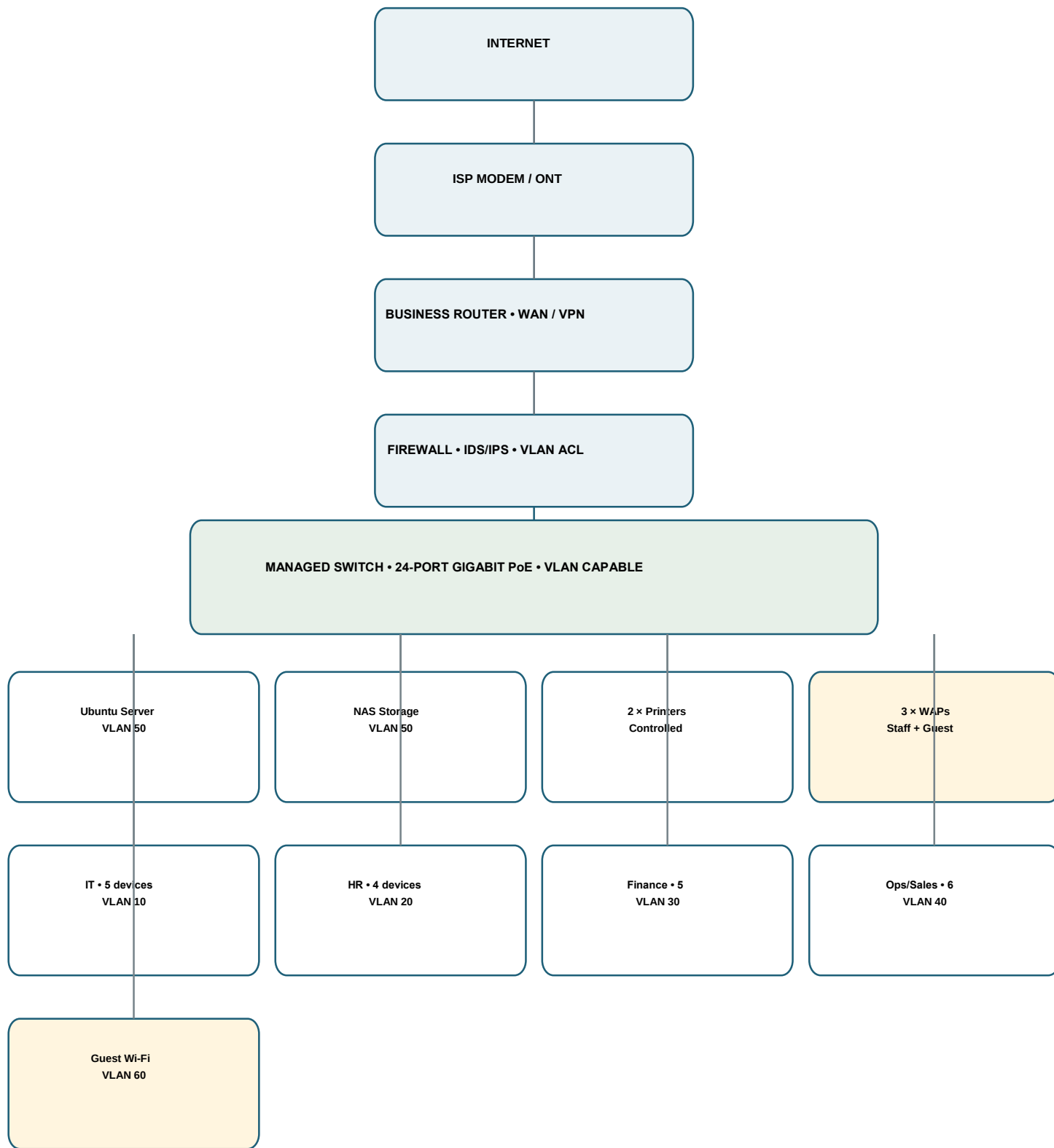
Asset ID	Network Equipment	Qty.	Purpose
NET-001	ISP Modem / ONT	1	Terminates the provider connection
NET-002	Business Router	1	Routes traffic between internal networks and the internet; supports VPN
NET-003	Firewall	1	Filters traffic, applies policies, and protects the internal network
NET-004	Managed Switch	1	Connects wired computers and devices; supports VLAN segmentation
NET-005	Wireless Access Point	3	Provides secure wireless access across the office
NET-006	Patch Panel	1	Organizes and terminates structured cable runs
NET-007	CAT6 Cable	2 boxes	Reliable Gigabit-capable wired network connections
NET-008	RJ45 Connectors	100 pcs	Cable termination, repairs, and spare connectors

Network Design Justification

All internet traffic passes through the ISP modem/ONT, router, and firewall before reaching the managed switch. The switch distributes wired connectivity to employee devices, server, NAS, printers, and APs. Recommended VLANs are: VLAN 10 IT, VLAN 20 HR, VLAN 30 Finance, VLAN 40 Operations/Sales, VLAN 50 Servers, and VLAN 60 Guest Wi-Fi. The guest VLAN should have internet-only access; firewall rules should limit access to HR and Finance records.

PART 5 — ENTERPRISE NETWORK DIAGRAM

Proposed Network Topology Diagram Notes



Logical design: Internet → ISP modem/ONT → business router → firewall → managed switch. Each server, storage, printer, access point, and department connects directly to the managed switch.

PART 6 — SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities: First-level employee support; resolves common hardware, software, account, printer, and application incidents; documents and escalates issues.

Skills: Troubleshooting, customer service, Windows, basic networking, communication, documentation.

Common Tools: Ticketing system, remote support, Windows utilities, device management, command-line tools.

Certifications: CompTIA A+, ITIL Foundation, Microsoft 365 Fundamentals.

6.2 Network Administrator

Responsibilities: Maintains routers, switches, Wi-Fi, VLANs, firewall rules, monitoring, and connectivity; troubleshoots outages and applies changes.

Skills: TCP/IP, routing and switching, IP addressing, wireless, security, monitoring.

Common Tools: Cisco IOS, Packet Tracer, Wireshark, ping, traceroute, monitoring platforms.

Certifications: CCNA; later CCNP Enterprise.

6.3 Linux System Administrator

Responsibilities: Installs, patches, hardens, and operates Linux servers; manages users, services, permissions, storage, logs, and backups.

Skills: Linux CLI, Bash, SSH, user/group management, networking, storage, security, troubleshooting.

Common Tools: Bash, SSH, systemctl, journalctl, apt, firewall tools, virtualization.

Certifications: RHCSA, Linux Foundation LFCS, CompTIA Linux+.

6.4 Cloud Administrator

Responsibilities: Manages cloud compute, storage, IAM, monitoring, costs, backups, deployments, and cloud security.

Skills: Cloud computing, networking, IAM, virtualization, monitoring, storage, automation.

Common Tools: AWS Console, AWS CLI, IAM, CloudWatch, cloud storage, virtual machines.

Certifications: AWS Certified Cloud Practitioner; then AWS SysOps Administrator.

How the Four Professionals Work Together

The Helpdesk Technician receives and documents user issues, resolves common cases, and escalates incidents with clear evidence. The Network Administrator handles connectivity, switching, Wi-Fi, and firewall problems. The Linux System Administrator keeps internal servers and services available, secure, patched, and backed up. The Cloud Administrator manages hosted systems, identity, storage, and cloud security. During incidents and planned changes, all four roles share documentation, follow change control, apply least privilege, and communicate the business impact and restoration status to management.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Subscribe to business-grade fiber service with a documented service-level agreement, responsive support, and at least 500 Mbps symmetrical bandwidth. Evaluate a secondary 5G/LTE or separate fiber line for failover.

Server Specifications

Start with an 8-core server CPU, 32–64 GB ECC RAM, two 1 TB enterprise SSDs in RAID 1 for the operating system, additional redundant data storage, dual Gigabit Ethernet, remote management, and UPS protection. Run Ubuntu Server LTS. RAID 1 improves availability for a drive failure but does not replace backups.

Backup Strategy

Follow the 3-2-1 model: retain three copies of important data on two types of storage with one copy offline or off-site.

Recommended backup locations:

- Production data on the server
- Scheduled backups to the NAS
- Rotating encrypted external drives with one stored separately

Test a file restore monthly and perform a documented recovery test at least twice a year.

Security Recommendations

Use the firewall with default-deny inbound rules, intrusion prevention, VPN, logging, and web filtering. Segment departments using VLANs; restrict HR and Finance records by role; use WPA3 Wi-Fi; isolate guest access; update systems and firmware; disable unused switch ports; use MFA; keep separate standard and administrator accounts; and conduct security awareness training.

Antivirus

Enable and keep Microsoft Defender updated on all Windows workstations. Use real-time protection, cloud-delivered protection, scheduled scanning, tamper protection, and centralized alert review when available. Do not install multiple real-time antivirus engines on one device.

Password Policy

Require unique passphrases of at least 14 characters, password-manager use, MFA for email, remote access, cloud services, and all administrator accounts. Prohibit account sharing and password reuse; use rate limiting/lockout; promptly remove access during offboarding; and issue separate named admin accounts.

Expansion Plan

Keep spare patch-panel and switch capacity, use structured CAT6 cabling, and select an expandable firewall/AP management platform. At 40 employees, add a managed switch and APs after a wireless survey, increase NAS capacity and backup storage, upgrade bandwidth if monitoring indicates saturation, and consider a second server or cloud-hosted services for resilience.

PART 8 — PERSONAL REFLECTION

This project helped me understand that system administration involves more than installing computers, configuring networks, or fixing technical problems. Before an IT infrastructure can be deployed, the needs of the organization must be analyzed and documented. Through this activity, I learned how to identify hardware, software, and networking requirements for a startup company based on its employee count and the responsibilities of each department.

The most challenging part for me was designing the enterprise network topology. I had to consider how the Internet, modem, router, firewall, switch, server, printer, wireless access points, NAS, and four departments should connect. I also needed to keep the design logical and easy to understand. This task showed me that network design requires careful planning because a missing component, incorrect connection, or poorly configured access rule can affect the entire organization.

I also learned the importance of inventories. A hardware, software, and network inventory lets an organization know what resources exist, where they are assigned, who supports them, and what they are used for. Proper documentation makes troubleshooting, maintenance, upgrades, purchasing, and future expansion easier. I now understand that security must be included from the start through firewalls, endpoint protection, strong authentication, patching, backups, and access controls.

Planning before deployment helps prevent unnecessary expense, compatibility problems, weak security, and poor allocation of resources. Instead of buying equipment randomly, a System Administrator should first understand both current and future business needs. Overall, this project will help me become a better System Administrator by improving my planning, networking, documentation, and decision-making skills. It also taught me that effective administrators communicate technical choices clearly and collaborate with other IT roles to keep business operations reliable and secure.

Overall Infrastructure Summary

Area	Key Plan
Company	20-employee BPO startup in Quezon City
Departments	IT (5), HR (4), Finance (5), Operations/Sales (6)
Core Hardware	20 desktops, 7 laptops, server, router, switch, printers, UPS, APs, NAS, backup drives, monitors
Network	ISP modem/ONT → router → firewall → managed switch; VLANs 10–60
Server	Ubuntu Server LTS; 8-core CPU; 32–64 GB ECC RAM; RAID 1
Backup	3-2-1 model; server + NAS + encrypted rotating external drives
Security	Firewall, VLAN segmentation, MFA, Defender, WPA3, least privilege, patching, backups
Growth	Designed for expansion to 40 employees with additional switching, APs, storage, bandwidth, and resilience

References

[1] Cisco, "CCNA 200-301 Exam Topics." Cisco identifies network fundamentals, network access, IP connectivity, IP services, security fundamentals, and automation/programming as core CCNA domains.

[2] Red Hat, "Red Hat Certified System Administrator." RHCSA includes essential command-line tools, services, storage, users, security, and basic container administration.

[3] AWS, "AWS Certified Cloud Practitioner." The certification validates foundational understanding of AWS Cloud, services, and terminology; its exam guide includes cloud concepts, security/compliance, services, and billing/pricing/support.

[4] Laguna State Polytechnic University, "ITEP 414 – System Administration and Maintenance, Week 2 Portfolio Project" (course module provided by user).

Document Basis: Reformatted from the supplied Ewan Co. Enterprise Infrastructure Plan while matching the organization, visual hierarchy, tables, and concise document flow of the supplied reference Enterprise Infrastructure Planning document.